

WARCOP RANGE, United Kingdom

WMO: 032260

Lat: 54.5725N

Lon: 2.4131W

Elev: 745

StdP: 14.30

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.2	26.3	20.3	15.6	27.7	22.9	17.7	29.9	41.4	38.5	36.0	38.9	2.8	140	0.683

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.3	73.7	64.1	70.2	61.9	67.2	60.2	65.5	71.2	63.1	68.5	61.2	65.7	10.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.3	89.6	67.5	61.0	82.3	65.4	59.2	77.3	63.5	30.7	71.6	28.9	68.6	27.6	65.7	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 33.8	28.6	24.5	DB	18.9	78.7	2.8	3.4	16.8	81.2	15.2	83.2	13.6	85.1	11.5	87.6	(4)
(5)			WB	18.6	70.7	2.8	4.7	16.6	74.1	15.0	76.8	13.5	79.4	11.4	82.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	47.3	38.3	38.6	41.0	44.9	49.9	54.7	58.3	57.6	53.8	48.4	42.8	38.6	(6)
(7)		DBStd	8.87	5.35	5.35	5.15	5.13	5.34	4.54	4.42	4.30	4.35	5.26	5.67	6.24	(7)
(8)		HDD50	1886	362	319	282	167	67	8	0	1	13	90	222	355	(8)
(9)		HDD65	6481	826	739	745	604	467	312	216	236	336	514	667	819	(9)
(10)		CDD50	897	0	1	2	14	65	148	257	236	127	41	6	0	(10)
(11)		CDD65	15	0	0	0	0	0	2	7	5	1	0	0	0	(11)
(12)		CDH74	84	0	0	0	0	1	10	41	31	1	0	0	0	(12)
(13)		CDH80	5	0	0	0	0	0	0	3	2	0	0	0	0	(13)
(14)	Wind	WSAvg	10.1	11.4	11.3	11.1	11.2	11.3	9.4	8.2	8.3	8.9	10.3	10.0	10.1	(14)
(15)	Precipitation	PrecAvg	53.8	6.3	4.8	3.8	3.2	3.2	3.1	3.5	4.0	4.1	5.0	5.8	6.8	(15)
(16)		PrecMax	66.9	10.0	11.7	7.2	5.9	5.9	7.0	7.9	9.6	7.5	9.8	14.5	17.7	(16)
(17)		PrecMin	39.3	1.3	0.9	1.1	1.5	0.9	1.0	1.9	0.9	1.0	1.9	2.7	2.1	(17)
(18)		PrecStd	7.0	2.4	3.0	1.3	1.2	1.4	1.5	1.3	1.8	1.5	2.0	2.6	3.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.7	53.5	59.2	65.9	72.3	75.0	78.4	78.3	70.7	63.5	56.6	52.5	(19)
(20)			MCWB	49.1	48.3	48.6	54.3	62.8	64.4	66.3	65.5	64.2	58.1	54.2	50.7	(20)
(21)		2%	DB	49.2	49.9	53.7	60.3	67.7	70.0	74.2	72.3	66.2	59.9	54.5	50.6	(21)
(22)			MCWB	47.3	47.1	47.7	51.5	59.5	61.0	64.1	64.0	60.6	56.2	52.4	48.9	(22)
(23)		5%	DB	47.6	48.1	50.6	56.0	63.3	66.5	70.6	67.9	63.1	57.9	52.5	49.0	(23)
(24)			MCWB	45.7	45.8	46.6	49.3	55.9	59.0	62.3	61.2	58.1	54.9	50.4	47.2	(24)
(25)		10%	DB	46.0	46.2	48.4	53.2	59.3	63.3	66.9	64.9	60.9	56.1	50.7	47.2	(25)
(26)			MCWB	44.2	44.2	45.3	47.8	53.0	56.6	60.4	59.7	56.7	53.3	48.6	45.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.9	50.1	51.7	55.8	66.4	65.5	70.7	69.3	65.6	59.4	55.0	51.3	(27)
(28)			MDCB	51.2	51.9	56.5	63.9	69.8	71.6	73.8	71.5	69.0	61.7	56.3	52.2	(28)
(29)		2%	WB	47.7	47.9	49.0	52.5	60.2	62.1	65.6	64.4	61.9	57.2	52.9	49.4	(29)
(30)			MDCB	48.9	49.5	52.0	57.9	66.3	68.8	72.3	69.5	64.9	59.0	54.2	50.5	(30)
(31)		5%	WB	46.1	46.2	47.3	50.4	56.6	59.6	63.1	62.3	59.4	55.4	50.8	47.5	(31)
(32)			MDCB	47.3	47.8	49.8	54.7	62.1	65.2	69.1	66.8	61.9	57.5	52.2	48.9	(32)
(33)		10%	WB	44.4	44.5	45.8	48.6	53.9	57.5	61.0	60.5	57.3	53.6	48.8	45.7	(33)
(34)			MDCB	45.9	46.0	48.1	52.6	58.4	62.3	65.7	64.1	60.0	55.8	50.5	47.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.6	8.8	10.9	12.7	13.7	13.2	13.3	12.3	11.8	10.2	8.6	8.4	(35)
(36)			MCDBR	8.9	9.9	13.9	17.3	19.8	18.8	19.6	17.2	14.4	11.0	8.7	9.4	(36)
(37)			MCWBR	9.0	8.7	9.8	11.5	13.4	12.3	12.3	10.9	10.0	8.8	8.4	9.3	(37)
(38)		5% WB	MCDBR	8.8	9.5	12.2	15.0	17.7	17.0	16.9	14.3	12.7	10.5	8.7	9.3	(38)
(39)			MCWBR	9.1	8.9	9.6	10.6	13.0	12.1	12.0	10.5	10.3	9.4	8.7	9.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.318	0.331	0.361	0.383	0.385	0.384	0.384	0.372	0.355	0.339	0.314	0.302	(40)	
(41)		taud	2.262	2.337	2.276	2.241	2.281	2.313	2.316	2.363	2.414	2.413	2.336	2.242	(41)	
(42)		Ebn at Noon	193	238	256	266	272	274	271	267	257	233	196	172	(42)	
(43)		Edh at Noon	19	24	32	38	39	38	37	34	28	23	17	16	(43)	
(44)	All-Sky Solar Radiation	RadAvg	176	388	741	1182	1534	1546	1494	1205	855	458	224	135	(44)	
(45)		RadStd	23	44	71	117	115	151	154	88	75	54	23	13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page